Finite-Horizon MDPs

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Deterministic Optimization

$$\max_{a} \; -\frac{1}{2} a^\top R a + a^\top M s + \frac{1}{2} s^\top Q s$$

- Assume $R > 0$. Thus, we have a (strictly) concave optimization problem
- The optimal solution is given by $-Ra + Ms = 0$ or $a = R^{-1}Ms$
- What if we changed the optimization problem to

$$\max_{\pi} -\frac{1}{2} \pi(s)^\top R \pi(s) + \pi(s)^\top M s + \frac{1}{2} s^\top Q s \,,$$

  i.e., maximization is over functions of s?

Solution is the same: Why?

- Suppose the optimal solution to the second problem is $\pi^*(s) \neq R^{-1}Ms$

- If we denote the objective in the first problem as $J(a)$, then the above statement implies $J\left(\pi^*(s)\right) > J(R^{-1}Ms)$ and $a^* = \pi^*(s)$ yields a lower value for the objective than $R^{-1}Ms$

- Thus $R^{-1}Ms$ cannot be the optimal solution, which is a contradiction since we know that due to the strict convexity of the problem, first-order necessary conditions yield unique optimal solutions

Stochastic Optimization

- As in the case of determaxistic optimization

$$\max_{\pi} E(r(s, \pi(s))$$

  can be solved by solving

$$\max_{a} r(s, a)$$

  - If there is a solution to the second problem, it will be a function of s
  - The equivalence between the two problems, i.e.,

$$\max_{\pi} E(r(s, \pi(s)) = E\left(\max_{a} r(s, a)\right),$$

    can be shown by contradiction as before

Optimization with Imperfect Observations

$$\max_{\pi} E(r(s, \pi(y))$$

- Here, we are optimizing over functions of y. It is assumed that we know the joint distribution of (s, y)

- Like the perfect information case, this problem can be solved by solving

$$\max_{a} E(\, r(s, a)|y)$$

Controlled Markov Processes

$$\max_{\pi_0,\ldots,\pi_{T-1}} \sum_{k=0}^{T-1} E\left(r(s_k, a_k)\right),$$

where

$$s_{k+1} = f(s_k, a_k, w_k)$$

and

$$a_k = \pi_k(s_0, a_0, .., s_{k-1,}\, a_{k-1}, s_k)$$

- w_k are independent
- a_k is the control action which is chosen as a function of the past history
- π_k is called the control policy

MDPs

$$\max_{\pi_0,\ldots,\pi_{T-1}} \sum_{k=0}^{T-1} E(r(s_k, a_k)),$$

where

$$P(s_{k+1} = j | s_k = i, a_k = a) = P(j|i,a) \text{ or } P_{ij}(a)$$

and

$$a_k = \pi_k(s_0, a_0, .., s_{k-1,} a_{k-1}, s_k)$$

- w_k are independent
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Markov Policies

- In the previous slide, we allowed the actions to depend on the entire history up to the current time slot
  - Note: the policy could be a randomized policy, i.e., it specifies a probability distribution over the allowed control actions
- A seemingly more restrictive set of policies are the so-called Markov policies which only depend on the current state, i.e., $a_k = \pi_k(s_k)$
  - As with policies that depend on the entire history, Markov policies may also be randomized policies
- It turns out that optimizing over the set of Markov policies is sufficient
  - We will provide an outline of the proof for countable state and action spaces

Outline of Proof

- Let $\pi = (\pi_0, \pi_1, \ldots, \pi_T)$ be a history dependent policy which induces a distribution p_k on (s_k, a_k)
  - i.e., $p_k(s_k, a_k)$ is the probability of the state and control being (s_k, a_k) in time k.
  - Abusing notation, we will use $p_k(s_k)$ to denote the marginal probability that the state is s_k at time k
- Let $\tilde{\pi}_k$ be a Markov policy which selects control action a_k with probability

$$\tilde{\pi}_k(a_k|s_k) = \frac{p_k(s_k, a_k)}{p_k(s_k)}$$

Outline of Proof (Continued)

- Let $\tilde{p}_k(s_k, a_k)$ be the probability distribution induced by the $\tilde{\pi}_k$ on (state, control)

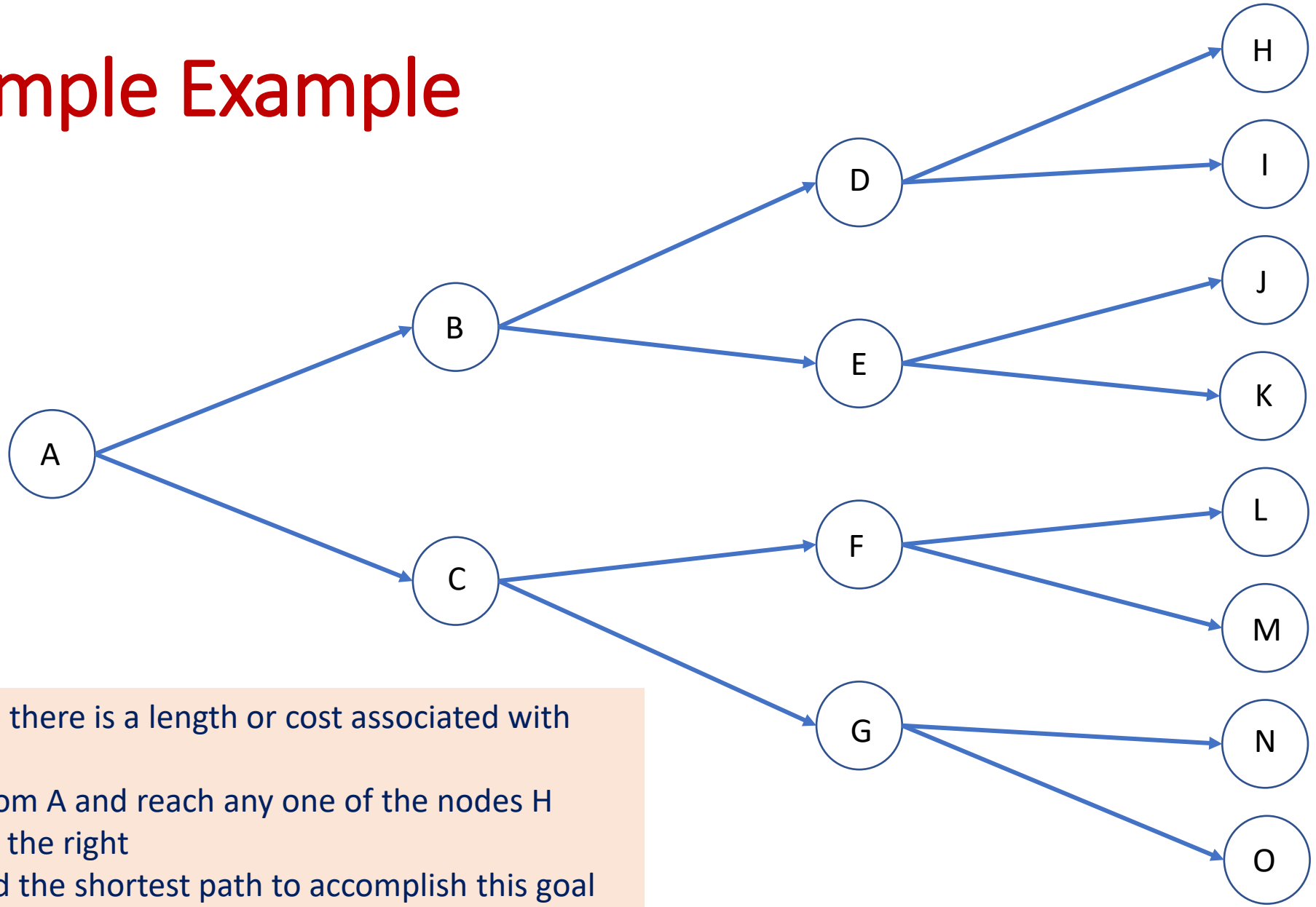
- The proof is completed by inductively showing that

$$p_k(s_k, a_k) = \tilde{p}_k(s_k, a_k) \; \forall s_k, a_k$$

  using the fact that $p_0(s_0) = \tilde{p}_0(s_0)$ since that is just the initial distribution of the state

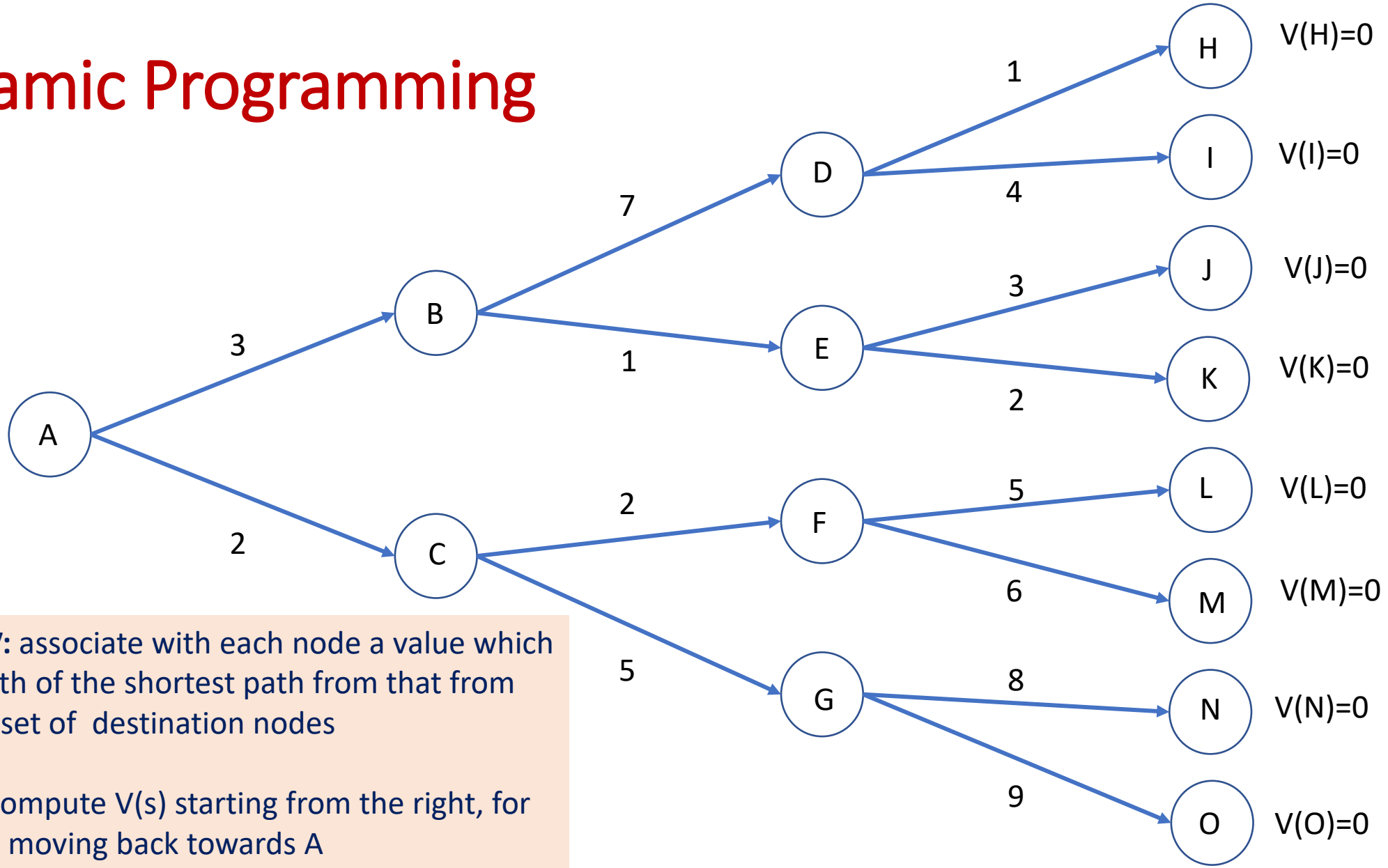
Deterministic Dynamic Programming

A Simple Example



Cost: Assume there is a length or cost associated with each edge
Goal: Start from A and reach any one of the nodes H through O on the right
Problem: Find the shortest path to accomplish this goal

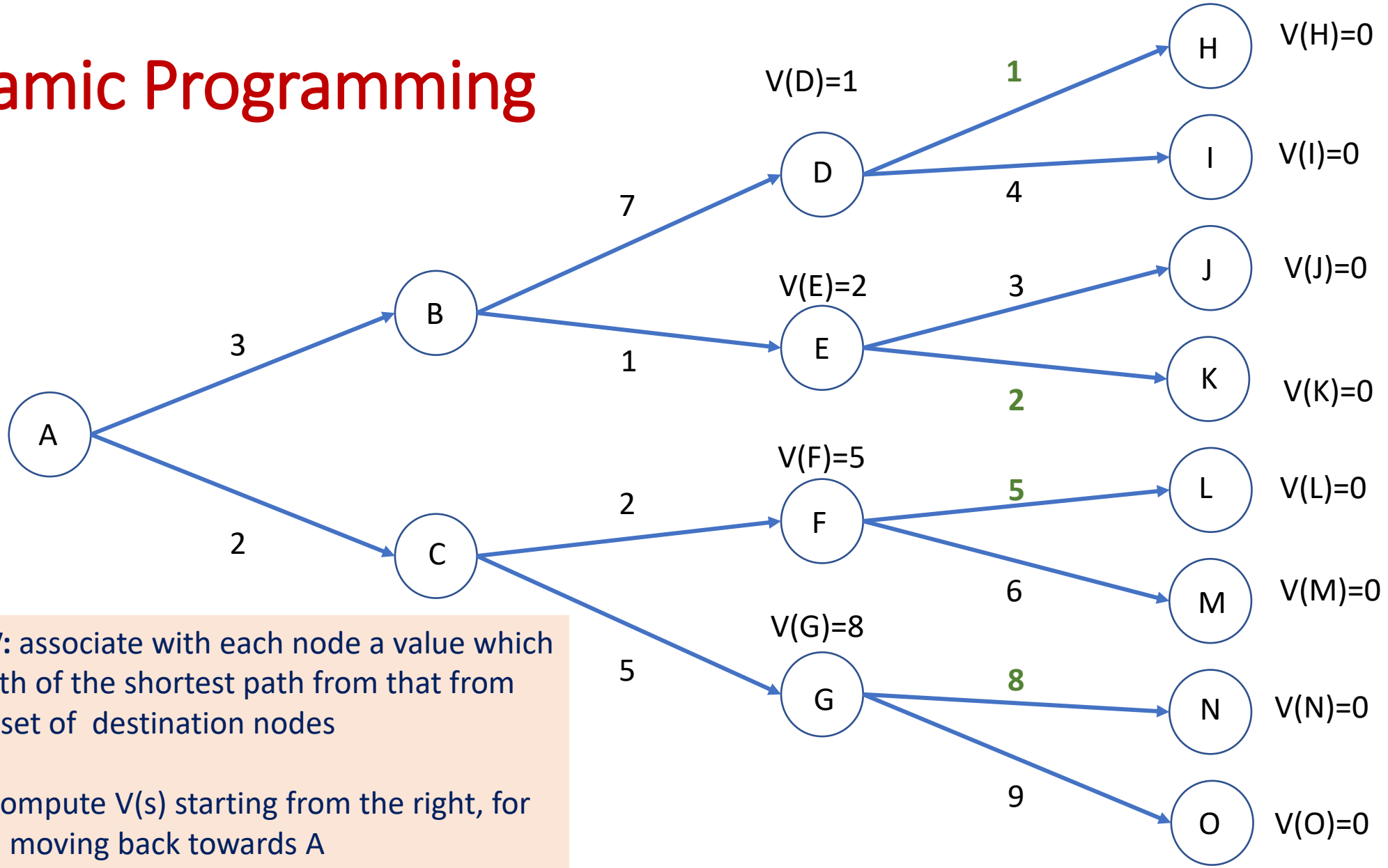
Dynamic Programming



Value Function V: associate with each node a value which denotes the length of the shortest path from that from that node to the set of destination nodes

DP: Recursively compute V(s) starting from the right, for each node s, and moving back towards A

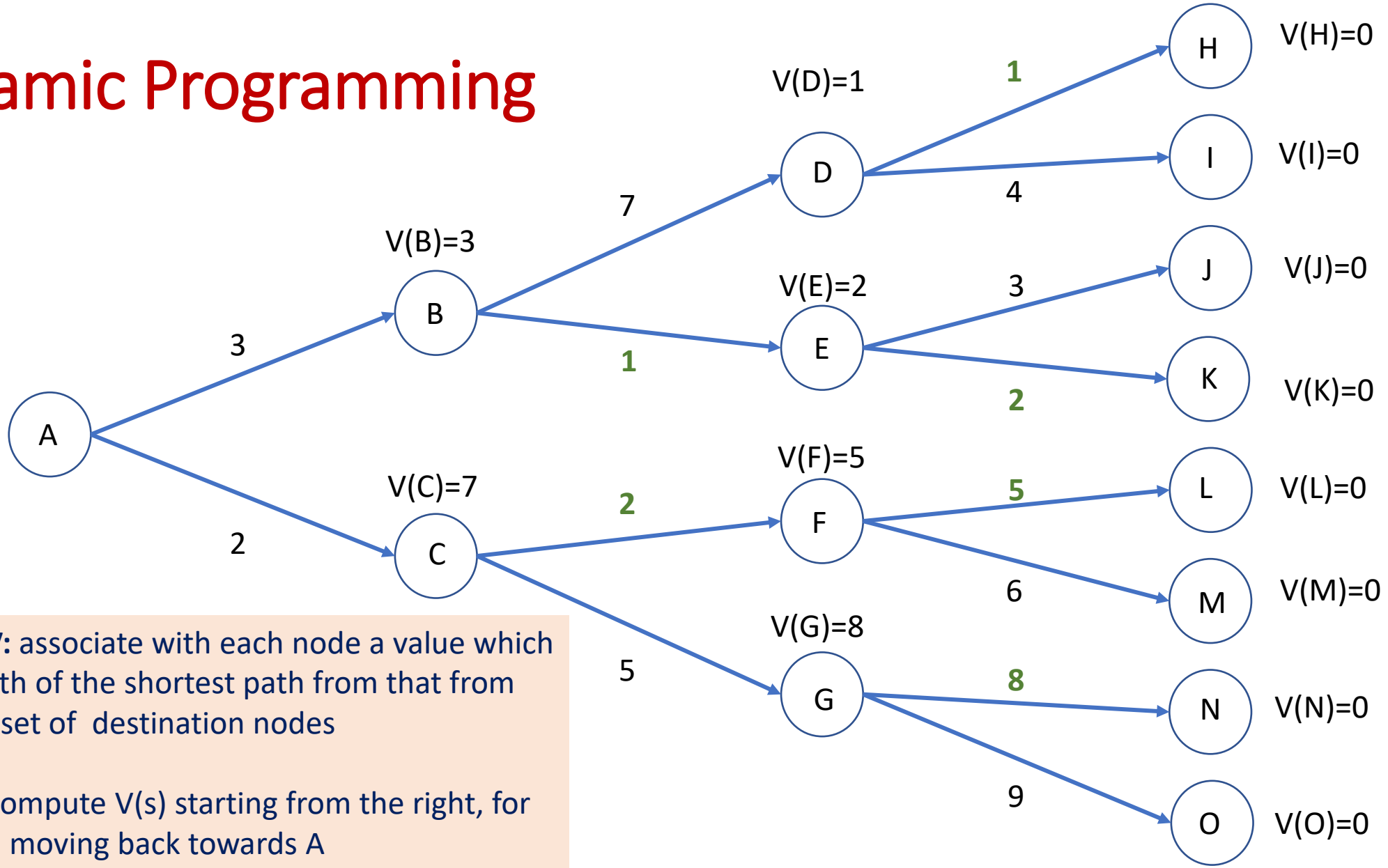
Dynamic Programming



Value Function V: associate with each node a value which denotes the length of the shortest path from that from that node to the set of destination nodes

DP: Recursively compute V(s) starting from the right, for each node x, and moving back towards A

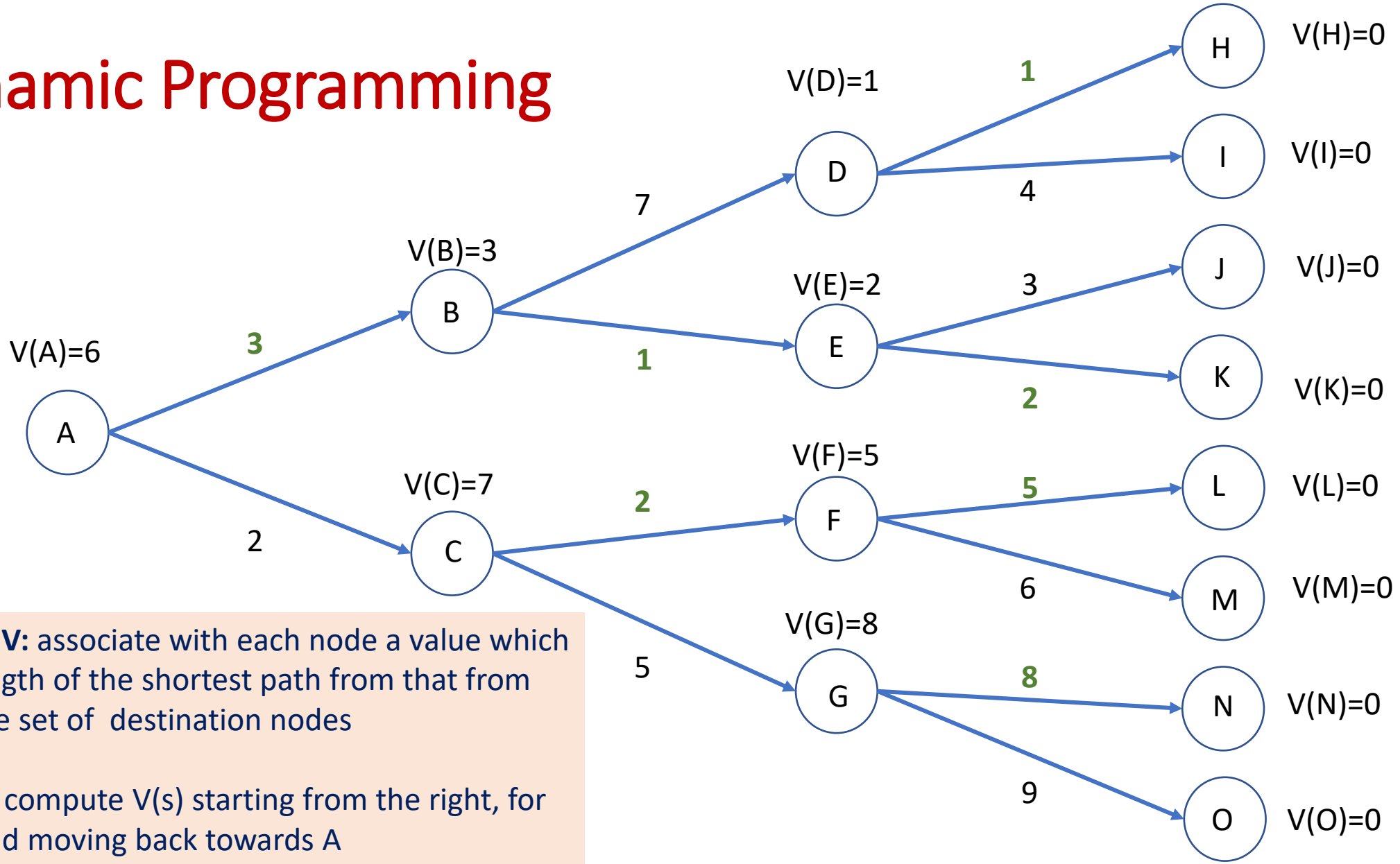
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Dynamic Programming

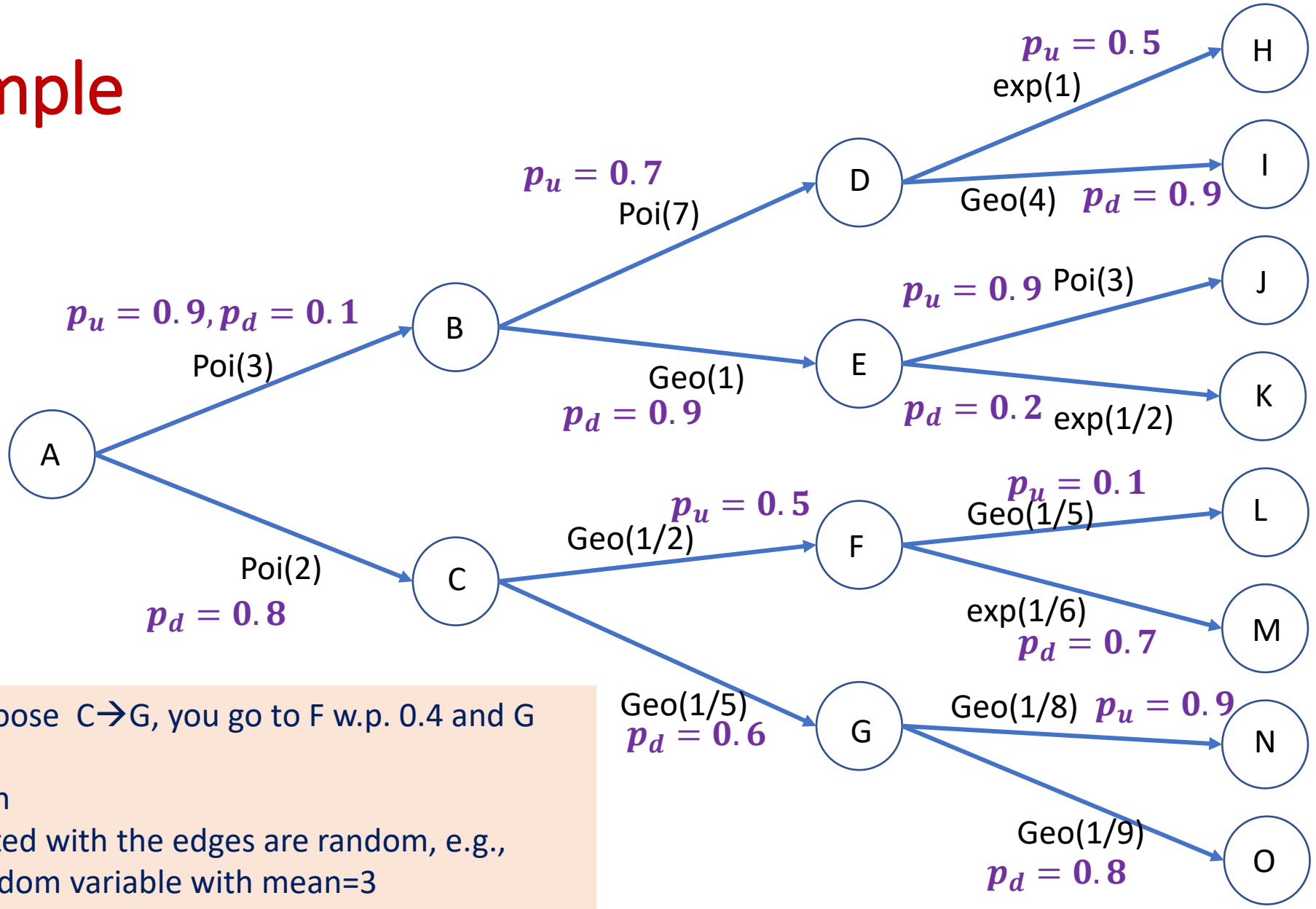


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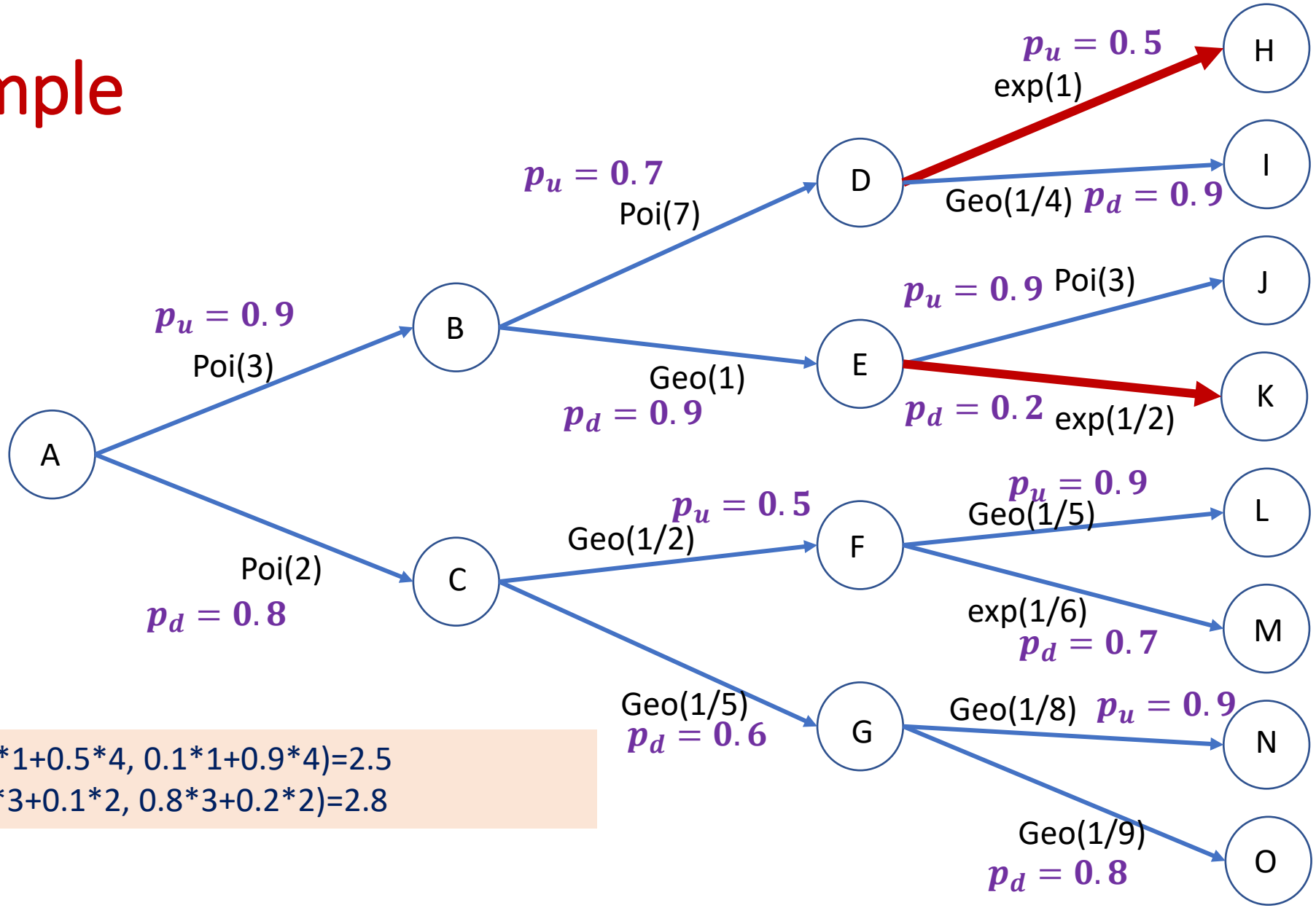
DP for Stochastic Systems

Example



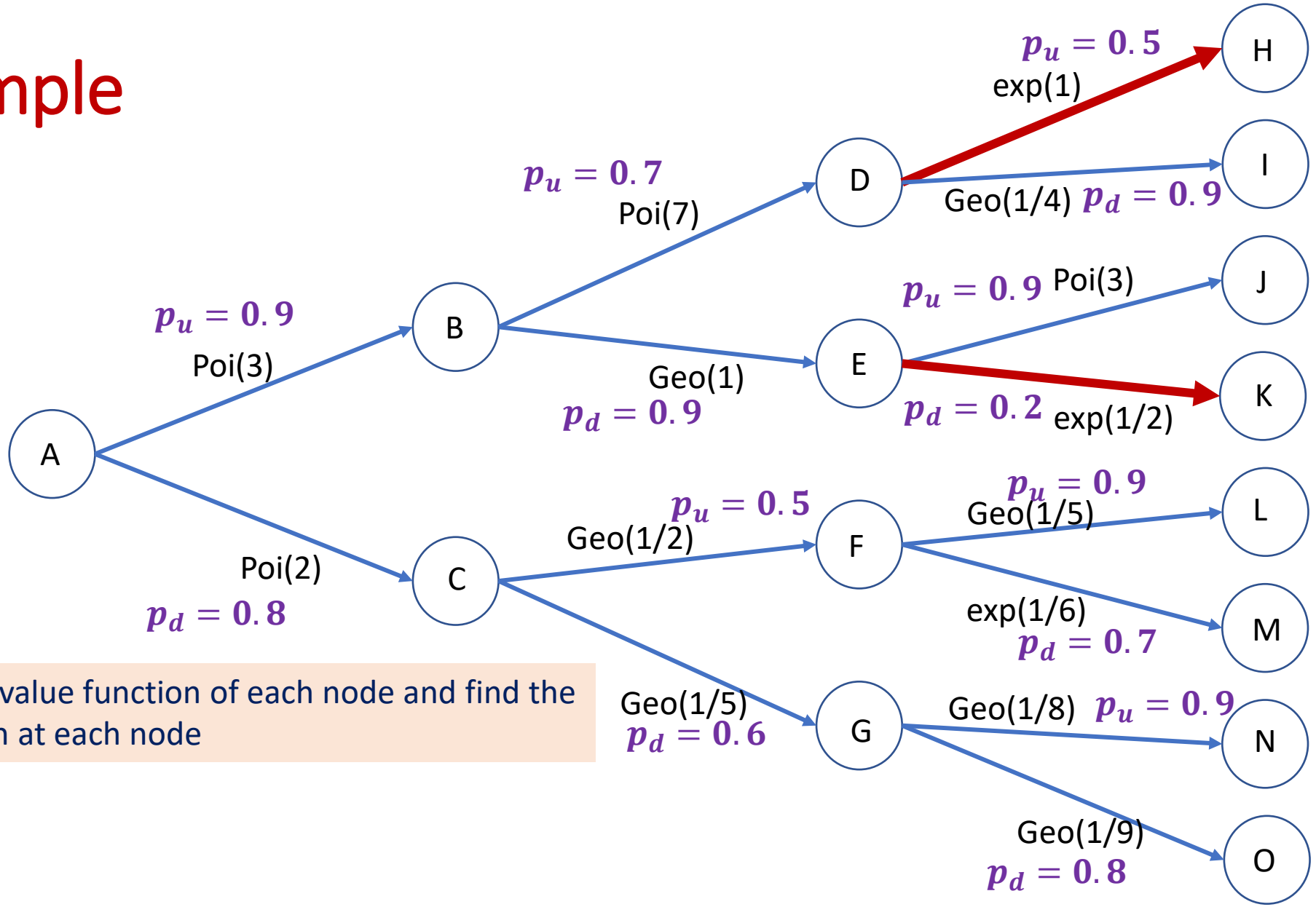
- When you choose C→G, you go to F w.p. 0.4 and G w.p. 0.6
- u: up, d: down
- Costs associated with the edges are random, e.g., Poi(3) is a random variable with mean=3
- Goal: Find a policy with smallest expected path length

Example



- V(D)=min(0.5*1+0.5*4, 0.1*1+0.9*4)=2.5
- V(E)=min(0.9*3+0.1*2, 0.8*3+0.2*2)=2.8

Example



- Compute the value function of each node and find the optimal action at each node

Controlled Stochastic Dynamical System

$$s_{k+1} = f_k(s_k, a_k, w_k)$$

- $\{w_k\}$ is an independent noise sequence
- The control action a_k can be a randomized action
- So $\{s_k\}$ is a controlled Markov process
- Cost: $r_k(s_k, u_k)$ (this could be expected cost conditioned on x_k, u_k)

- Problem: Find a policy μ_k such that $u_k = \mu_k(x_k)$ to solve

$$\max E\left(\sum_{k=0}^{N-1} r_k(s_k, a_k) + r_N(s_N)\right)$$

DP equation

- At stage k, if you are in state x_k, the optimal control u_k is obtained by solving

$$V_k(s) = \max_u r_k(s,u) + E(V_{k+1}(s_{k+1})|s_k = s, a_k = a)$$

where

$$s_{k+1} = f(s_k, a, w_k),$$

with the value function initialized as

$$V_N(s_N) = r_N(s_N)$$

MDPs

- s_k takes on values in a finite set

- The dynamics can be described by the transition probability

$$Prob(s_{k+1} = j | s_k = i, a_k = a)$$

- If the transition probability does not explicitly depend on k, then we have

$$P_{ij}(u) \triangleq Prob(s_{k+1} = j | s_k = i, s = u)$$

DP or Bellman Equation

$$V_k(i) = \max_u r(i,a) + E(V_{k+1}(x_{k+1})|x_k = i, a_k = a)$$
$$= \max_u r(i,a) + \sum_j P_{ij}(a) V_{k+1}(j)$$

with

$$V_N(i) = r_N(i)$$

- The optimal solution is a deterministic Markov policy!

Example: Exercising a Stock Option

- You have the option of buying a stock for P dollars within N days
  - The current price of the stock is x_0
- The stock price on day k is x_k which satisfies
  - $x_{k+1} = (1+\epsilon)x_k$ w.p. p and $= (1-\delta)x_k$ w.p. $1-p$
  - $\epsilon > 0, 0 < \delta < 1$
- Let u_k indicate whether the option is exercised after observing the stock price on day k.

DP Solution

- $V_N(x_N) = \max(x_N - P, 0) = (x_N - P)^+$, where 0 corresponds to not exercising the stock option

- $V_k(x_k) = \max\{x_k - P, pV_{k+1}\big((1+\epsilon)x_k\big) + (1-p)V_{k+1}((1-\delta)x_k)\}$
  - Thus, the option is exercised on day k if the stock price exceeds $T_k(x) = P + pV_{k+1}\big((1+\epsilon)x_k\big) + (1-p)V_{k+1}((1-\delta)x_k)$
  - The threshold $T_k(x)$ for exercising the option can be computed numerically if p, ϵ, δ are known.

Example: The Hiring Problem

- You want to hire one software engineer
  - There are N candidates.
  - After interviewing each candidate, you have to either decide to hire them or reject them permanently.
    - The challenge is to make a decision based on previous candidates without looking at future candidates.
  - The goal is to maximize the probability of picking the best candidate

Model

- After interviewing k candidates, the probability the next one is i^{th} best candidate $(i \leq k+1)$ that you have interviewed so far is $\frac{1}{k+1}$
  - In other words, the ranking of each new candidate is picked uniformly among all the candidates so far
  - Given that the kth one is the best so far, the probability that they are the best overall is $\frac{k}{N}$. This is because the probability that the next one is ranked lower than k is $\frac{k}{k+1}$, the probability that one after that is ranked lower than $\frac{k+1}{k+2}$. So the probability that the kth is the best is given by $\frac{k}{k+1} * \frac{k+1}{k+2} * \cdots * \frac{N-1}{N}$.

MDP Formulation

- s_k: the state of the system is k itself, i.e., the index of the candidate who was the last one to be interviewed. Thus, the state takes values $0,1,2,\ldots,N-1$
- Therefore, DP gives

$$V(k) = \frac{k}{k+1}V(k+1) + \frac{1}{k+1}\max\left(\frac{k+1}{N}, V(k+1)\right)$$

  - $\frac{k}{k+1}$ is the probability that k^{th} candidate is not the best so far
  - $\frac{1}{k+1}$ is the probability they are the best so far and if you pick them, the probability that you picked the best candidate among all N is $\frac{k+1}{N}$
  - Note: $V(N) = 0$

Bellman Equatiom

$$V(k) = \frac{k}{k+1}V(k+1) + \frac{1}{k+1}\max\left(\frac{k+1}{N}, V(k+1)\right) = \max\left(\frac{k}{k+1}V(k+1) + \frac{1}{N}, V(k+1)\right)$$

- V is a non-increasing function, from the above equation from the second expression
- From the first expression, since $\frac{k}{N}$ is increasing in k and $V(k)$ is non-increasing, there exists a smallest τ such that $\frac{k}{N} \geq V(k)$, and i.e., if $k \leq \tau$, you don't select the candidate; if $k > \tau$, you select if they are the best so far and

$$V(k) = \frac{k}{k+1}V(k+1) + \frac{1}{N}, \qquad V(N) = 0$$

The Solution

$$V(k) = \frac{k}{k+1}V(k+1) + \frac{1}{N}, k \geq \tau; \quad V(N) = 0$$

- Solving,

$$V(k) = \frac{k}{N}\sum_{t=k}^{N-1}\frac{1}{t}$$

- Thus, τ is the smallest positive integer s.t. $V(\tau) \leq \frac{\tau}{N}$, or

$$\sum_{i=\tau}^{N}\frac{1}{i} \leq 1$$

Asymptotics for optimal k^*

It is the first candidate such that

$$1 \geq \sum_{i=\tau}^{N} \frac{1}{i} \approx \int_{\tau}^{N} \frac{1}{x} dx = \log \frac{N}{\tau} ,$$

Or

$$\frac{\tau}{N} \geq \frac{1}{e}$$

- Asymptotic Solution: pick the next best candidate so far after you have seen at least $\frac{1}{e}$ fraction of the candidates